

Spring 2022.

Q1.

Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8}$ (i)

$$= \lim_{x \rightarrow +\infty} \frac{\sqrt{x^4(3 + \frac{1}{x^3})}}{x^2(1 - \frac{8}{x^2})}$$

$$= \lim_{x \rightarrow +\infty} \frac{x^2 \sqrt{3 + (\frac{1}{x^3})}}{x^2(1 - \frac{8}{x^2})}$$

$$= \frac{\sqrt{3 + \frac{1}{\infty}}}{1 - \frac{8}{\infty}} = \frac{\sqrt{3 + 0}}{1 - 0}$$

$$= \sqrt{3}$$

(ii)

$y = x^5 \sec(\frac{1}{x})$; find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = 5x^4 \sec(\frac{1}{x}) + x^5 \sec(\frac{1}{x}) \tan(\frac{1}{x}) (\frac{1}{x^2})$$

$$= 5x^4 \sec(\frac{1}{x}) + x^3 \sec(\frac{1}{x}) \tan(\frac{1}{x})$$

$$= x^3 \sec(\frac{1}{x}) (5x - \tan(\frac{1}{x}))$$

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Confirm that formula $(5+\Delta x)^3$
 $\approx 125 + 75\Delta x$ is the local
approximation of $f(x) = (4+x)^3$
at $x_0 = 1$ where $\Delta x = x - 1$.

Solution:

$$f(x) = (4+x)^3$$

$$f'(x) = 3(4+x)^2 \cdot 1$$

$$f'(x) = 3(4+x)^2$$

$$f'(x) = 3(4+x)^2$$

where

$$x_0 = 1$$

Formula of linear approximation
is:

$$L(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$x_0 = 1 \text{ Put in } \dots$$

$$f'(1) = 3(4+1)^2$$

$$f'(2) = 3(5)^2 = 3(25)$$

$$f'(1) = 75 \quad \text{--- (i)}$$

Put $x_0 = 1$ in

$$f(1) = (4+1)^3$$

$$f(1) = 125 \quad \text{--- (ii)}$$

Putting Equ (i) and (ii) in formula.

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$$L(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$L(x) = 125 + 75(x - 1)$$

$\therefore (\Delta x = x - 1)$ is given in equation.

So

$$L(x) = 125 + 75(\Delta x)$$

$$L(x) = 12.5 + 75(\Delta x)$$

$$(4 + x^3) \approx 125 + 75\Delta x$$

Hence Proved.

(iv).

Estimate the area between the graph of function $f(x) = \frac{1}{x}$ and the interval $[a, b] = [1, 2]$.

Estimate the specified area using $n = 5$ rectangles.

left end point $= a = 1$.

right end point $= b = 2$

$$\Delta x = \frac{b - a}{n} = \frac{2 - 1}{5} = \frac{1}{5}$$

$$a + \frac{\Delta x}{5} = 1.2$$

1. $[1, 1.2]$

2. $[1.2, 1.4]$

3. $[1.4, 1.6]$

4. $[1.6, 1.8]$

5. $[1.8, 2]$

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let evaluate functions $f(x) = \frac{1}{x}$ at these points.

$$1. f(1.2) = \frac{1}{1.2} \approx 0.833.$$

$$2. f(1.4) = \frac{1}{1.4} \approx 0.7143.$$

$$3. f(1.6) = \frac{1}{1.6} \approx 0.625$$

$$4. f(1.8) = \frac{1}{1.8} \approx 0.5556$$

$$5. f(2) = \frac{1}{2} = 0.5$$

$$\begin{aligned} \text{Riemann sum} &= \Delta x \cdot (f(x_1) + f(x_2) + \dots + f(x_5)) \\ &= \frac{1}{5} (0.833 + 0.7143 + 0.625 + 0.5556 + 0.5) \end{aligned}$$

$$= \frac{1}{5} (3.2282).$$

$$\text{Riemann sum} \approx 0.64564.$$

$$\int_0^{\sqrt{\pi}} 5x \cos(x^2) dx.$$

$$U = x^2 \Rightarrow dU = 2x dx \Rightarrow \frac{dU}{2} = dx.$$

$$x = 0 \rightarrow U = 0.$$

$$x = \sqrt{\pi} \rightarrow U = \pi$$

$$\int_0^{\sqrt{\pi}} 5x \cos(x^2) dx = \frac{5}{2} \int_0^{\pi} \cos U dU$$

$$= \frac{5}{2} (\sin U)_0^{\pi}$$

$$= \frac{5}{2} [0 - 0]$$

$$= 0.$$

examine

(9, 2, 0),

(0, -5, -3)

same

Condition.

Formula:

$|AB| =$

$|x_1 - x_2| + |y_1 - y_2|$

$$)^2 + (2 - 9)^2 + (0 - 7)^2$$

$$9 + 49 + 49$$

$$(1 - 9)^2 + (-5 - 2)^2 + (-3 - 0)^2$$

$$81 + 49 +$$



$$\sqrt{a^2 + b^2} + 1 \text{ يا } 3-7$$

له ٲٲسك

ٲٲ 2

ٲٲ $\frac{p_3}{p_1} \neq \frac{p_2}{p_1}$

Therefore,

Same



Q3

Evaluate the integral.

$$\int_0^3 \frac{x^3}{(3+x^2)^{5/2}} dx.$$

$$\text{let } t = 3+x^2. \quad x^2 = (t-3)$$

$$dt = 2x dx.$$

$$\frac{dt}{2} = x dx.$$

$$\int_0^3 \frac{x^2 \cdot x dx}{(3+x^2)^{5/2}} = \int_0^3 \frac{(t-3) dt}{t^{5/2} \cdot 2}$$

$$= \frac{1}{2} \int_0^3 \frac{(t-3)}{t^{5/2}} dt.$$

By separating both equations.

$$= \frac{1}{2} \left[\int_0^3 t^{-3/2} dt - 3 \int_0^3 t^{-5/2} dt \right]$$

$$= \frac{1}{2} \left[\frac{t^{-1/2}}{-1/2} - 3 \frac{t^{-3/2}}{-3/2} \right]_0^3$$

$$= \frac{+2}{2} [t^{-3/2} - t^{-1/2}] = \left[\frac{1}{t\sqrt{t}} - \frac{1}{\sqrt{t}} \right]$$

$$= \left[\frac{1-t}{t\sqrt{t}} \right] = \left[\frac{1-(3+x^2)}{3+x^2(\sqrt{3+x^2})} \right]$$

$$= \left[\frac{1-3+x^2}{3+x^2\sqrt{3+x^2}} \right]_0^3 = \left[\frac{-2+x^2}{3+x^2\sqrt{3+x^2}} \right]_0^3$$

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Applying limits

$$= \left[\frac{-2+x^2}{(3+x)\sqrt{3+x^2}} \right]_0^3$$

$$= \left[\frac{-2+(3)^2}{(3+3)\sqrt{3+3^2}} - \frac{-2+0}{(3+0)\sqrt{3}} \right]$$

$$= \left[\frac{-2+9}{12 \cdot 2\sqrt{3}} + \frac{2}{3\sqrt{3}} \right]$$

$$= \left[\frac{-2+9}{8 \times 3\sqrt{3}} + \frac{2}{3\sqrt{3}} \right]$$

$$= \left[\frac{7}{8 \times 3\sqrt{3}} + \frac{2}{3\sqrt{3}} \right]$$

$$= \left[\frac{7+2 \times 8}{8 \times 3\sqrt{3}} \right]$$

$$= \left[\frac{7+16}{24\sqrt{3}} \right] = \frac{23}{24\sqrt{3}}$$

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Find the distance between the skew lines $L_1: x = 3-t, y = 4+4t, z = 1+2t$ and $L_2: x = t, y = 3, z = 2t$.

$$L_1: x = 3-t, y = 4+4t, z = 1+2t.$$

$$L_2: x = t, y = 3, z = 2t.$$

$$Q, (3, 4, 1), P(0, 3, 0)$$

$$n = n_1 \times n_2 = \begin{vmatrix} i & j & k \\ 1 & 4 & 2 \\ 1 & 0 & 2 \end{vmatrix}$$

$$= i(8-0) - j(-2-2) + k(0-4) \\ = 8i + 4j - 4k \quad \text{--- (i)}$$

Using P to find equation.

$$8(x-0) + 4(y-3) - 4(z-0) = 0.$$

$$8(x-0) + 4y - 12 - 4z = 0.$$

$$8x + 4y - 4z - 12 = 0.$$

Now Also Using $Q(3, 4, 1)$ to find D.

$$D = \frac{|8(3) + 4(4) - 4(1) - 12|}{\sqrt{8^2 + 4^2 + (-4)^2 - 12^2}} = \frac{24 + 16 - 16}{\sqrt{64 + 16}} \\ = \frac{24}{\sqrt{208}}.$$